Module 10

Exotic Options and Structured Products

November 3 & 5, 2025

Exotic Options

Perpetual American Options:

- Perpetual American call on a dividend paying stock:

  At anytime to perpetuity the holder of the option may exercise to receive

  $payoff = \max(S-K,0)$

  BSM PDE:

  $$\frac{\partial f}{\partial t}+(r-q)S\frac{\partial f}{\partial S}+\frac{1}{2}\sigma^2S^2\frac{\partial^2 f}{\partial S^2}-rf=0 \ , \ payoff=\max(S-K,0)$$

  Assume r, q and σ are time independent → time independent solution

  → homogeneous linear ordinary differential equation (ODE):

  $$(r-q)S\frac{\partial f}{\partial S}+\frac{1}{2}\sigma^2S^2\frac{\partial^2 f}{\partial S^2}-rf=0$$

  General solution:

  $$f=AS^{\alpha^+}+BS^{\alpha^-}$$

  where

  $$\alpha^{\pm}=\frac{-\left(r-q-\frac{1}{2}\sigma^2\right)\pm\sqrt{\left(r-q-\frac{1}{2}\sigma^2\right)^2+2r\sigma^2}}{\sigma^2} \ ,$$

  are the solutions of

  $$\frac{1}{2}\sigma^2\alpha(\alpha-1)+(r-q)\alpha-r=0$$

  However $\alpha^-<0: \ S\to 0, f\to\infty$ → $B=0$

  → $C=AS^{\alpha^+}$

No arbitrage → $C \geq S - K$. At the exercise boundary, $S = S_*$

$$AS_*^{\alpha^+} = S_* - K \rightarrow A = \frac{S_* - K}{S_*^{\alpha^+}} \rightarrow C = \begin{cases} \frac{S_* - K}{S_*^{\alpha^+}} S^{\alpha^+} & S < S_* \\ S - K & S \geq S_* \end{cases}$$

Fixing S_* by maximize the option value:

$$\frac{\partial C}{\partial S_*} = 0 \rightarrow S_* = \frac{\alpha^+ K}{\alpha^+ - 1} \rightarrow C = \begin{cases} \frac{K}{\alpha^+ - 1}\left(\frac{\alpha^+ - 1}{\alpha^+}\frac{S}{K}\right)^{\alpha^+} & S < S_* \\ S - K & S \geq S_* \end{cases}$$

Smooth-pasting condition:

Maximizing American option ←→ continuity of option price and option delta at exercise boundary

Example in marketplace: perpetual warrants

- Perpetual convertible preferred stocks (CPS):

  Assume that f = CPS has notional N , pays coupon (dividend) at continuous rate Q . At any time (to perpetuity) a CPS holder may convert it into R_C shares (R_C = "conversion ratio")

  $$\Pi = f - mS, m = \frac{\partial f}{\partial S}, 0 \leq t < T$$

  $$\delta\Pi = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2}\right)\delta t + NQ\delta t - qS\frac{\partial f}{\partial S}\delta t$$

  $$\rightarrow \delta\Pi = \left(\frac{\partial f}{\partial t} - qS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} + NQ\right)\delta t = r\Pi\delta t = r\left(f - S\frac{\partial f}{\partial S}\right)\delta t$$

  $$\frac{\partial f}{\partial t} + (r-q)S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} - rf + NQ = 0$$

  Assume r , q , Q and σ are time independent, then perpetuity + homogeneity under constant parameters lead to

  $$(r-q)S\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} - rf + NQ = 0$$

  This is an inhomogeneous ODE.

Solution: general solution of the homogeneous ODE + a particular solution:

$$CPS = AS^{\alpha^+} + \frac{QN}{r} \ , \ \ CPS \geq R_C S$$

At the exercising boundary $S = S_*$:

$$AS_*^{\alpha^+} + \frac{QN}{r} = R_C S_* \ \ \rightarrow \ \ A = \frac{R_C S_* - \frac{QN}{r}}{S_*^{\alpha^+}}$$

$$\rightarrow \ CPS = \begin{cases} \dfrac{R_C S_* - \dfrac{QN}{r}}{S_*^{\alpha^+}} S^{\alpha^+} + \dfrac{QN}{r} & S < S_* \\ R_C S & S \geq S_* \end{cases}$$

Fixing S_* by maximizing the CPS value (or applying the smooth-pasting condition):

$$S_* = \frac{QN}{rR_C}\frac{\alpha^+}{\alpha^+ - 1} \ \rightarrow \ CPS = \begin{cases} \dfrac{\dfrac{QN}{r}}{\alpha^+ - 1}\left(\dfrac{\alpha^+ - 1}{\alpha^+} \dfrac{S}{\left(\dfrac{QN}{r}\right)} \right)^{\alpha^+} & S < S_* \\ R_C S & S \geq S_* \end{cases}$$

Perpetual CPS in marketplace:

- Discrete coupon (dividend)
- "non-cumulative" clause
- Call provision
- Put provision
- Contingent conversion ("CoCo") provision
- Credit risk

- Perpetual American put options:

  At anytime to perpetuity the holder of the option may exercise to receive

  $payoff = \max(K - S, 0)$

  Pricing: similar math as that for perpetual call applied to put payoff (HW, Hull)

Perpetual American puts in marketplace:

- Currently mostly traded OTC off-shore
- Common in cryptocurrency markets

Binary (digital) options:

- Payoff:
  - European binary call:
    - Cash call:

$$payoff = HI_K,\ H = \text{cash amount},\ I_K(S_T) = \begin{cases} 1 & S_T \geq K \\ 0 & S_T < K \end{cases}$$

    - Asset call:

$$payoff = S_T I_K$$

$$c(S_T, T) = \max(S_T - K, 0) = S_T I_K - K I_K$$

→ European call option = long binary asset call, short binary cash call

Cash call = $He^{-r(T-t)}N(d_2)$

Asset call = $Se^{-q(T-t)}N(d_1)$

Replication:

Difficult to replicate dynamically using underlying asset directly due to the discontinuity in the payoff

Let $b_c(K)$ = European cash binary with barrier K that pays one unit cash

$$b_c(K) = -\lim_{\eta \to 0} \frac{1}{\eta}\big(c(K+\eta) - c(K)\big)$$

$$= \lim_{\eta \to 0} \frac{1}{\eta}\big(c(K-\eta) - c(K)\big)$$

$$= -\lim_{\eta \to 0} \frac{1}{2\eta}\big(c(K+\eta) - c(K-\eta)\big)$$

$$= -\frac{\partial c(K)}{\partial K}$$

→ may be replicated as a limiting case of call spread

→ dealers may quote $b_c(K)$ as a call spread with a finite η.

→ Magnitude of η, form of the limit approximation depend on side of market, the shape of volatility skew, competitiveness of the market etc

- European binary put:
  - Cash put:

    $payoff = H(1 - I_K)$

  - Asset put:

    $payoff = S_T(1 - I_K)$

  $p(S_T, T) = \max(K - S_T, 0) = K(1 - I_K) - S_T(1 - I_K)$

  → European put option = long binary cash put, short binary asset put

  Cash put = $He^{-r(T-t)}N(-d_2)$

  Asset put = $Se^{-q(T-t)}N(-d_1)$

- American binaries:
  - One-touch binary call:

    $payoff = H\left(1 - I_{(0,K,t,T)}\right)$

  - No-touch binary call:

    $payoff = HI_{(0,K,t,T)}$

    $$I_{(K_1,K_2,t_1,t_2)}\{S_t\} = \begin{cases} 1 & K_1 < S_t < K_2, t_1 < t < t_2 \\ 0 & otherwise \end{cases}$$

  Closed form solutions may be obtained under constant parameters

  Parity:

  Prob(one touch) + Prob(no-touch) = 1

- Examples of usage:

Besides used as stand alone products, they can also be used in combinations:

- Principal protected notes
- Range options: e.g.,

  Let $K_1 < K_2$

  Payoff $= I_{(K_1,K_2)}(S_T) = \begin{cases} 1 & K_1 \le S_T < K_2 \\ 0 & otherwise \end{cases} = b_c(K_1) - b_c(K_2)$ ("digital spread")

- Range accruals: e.g.,

  For $0 = t_0 < t_1 < t_2 < ... < t_n = T$,

  $$Payoff = \max\left(floor, H\sum_{i=1}^{n} I_{(K_1,K_2,t_{i-1},t_i)} \right)$$

  Range accruals may have multi-asset underlying

- Embedded in other structured products …

- Risk:

  Consider a binary call for example:

  - Sensitivity to volatility:
    - S forward > K : increasing vol → decreasing likelihood of S end up ITM → negative vega
    - S forward < K : increasing vol → increasing likelihood of S end up ITM → positive vega
  - Sensitive to volatility skew:

    $$b_c(K) = -\frac{\partial c(K)}{\partial K} = e^{-r(T-t)}N(d_2) - \frac{\partial c(K)}{\partial \sigma}\frac{\partial \sigma(K)}{\partial K}$$

    = BSM + skew correction

    Presence of skew: $\frac{\partial \sigma(K)}{\partial K} < 0$ → skew makes binary more expensive

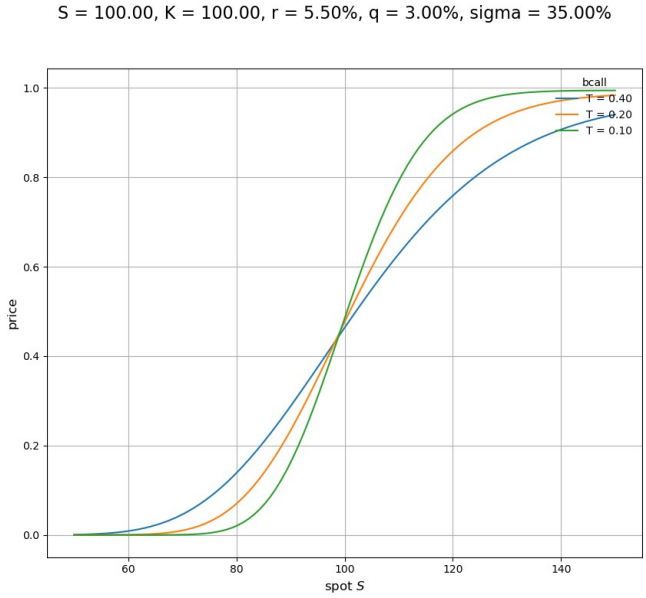
    Selling binary → selling skew

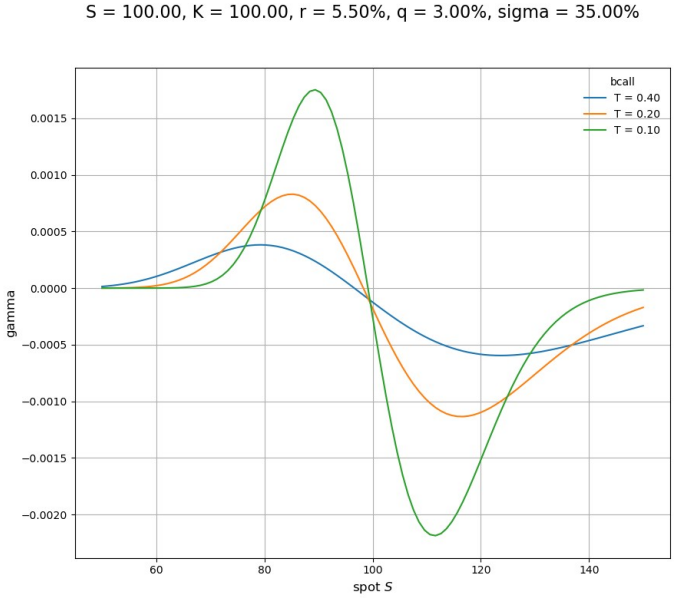
    American binaries: time dimension of volatility is also important

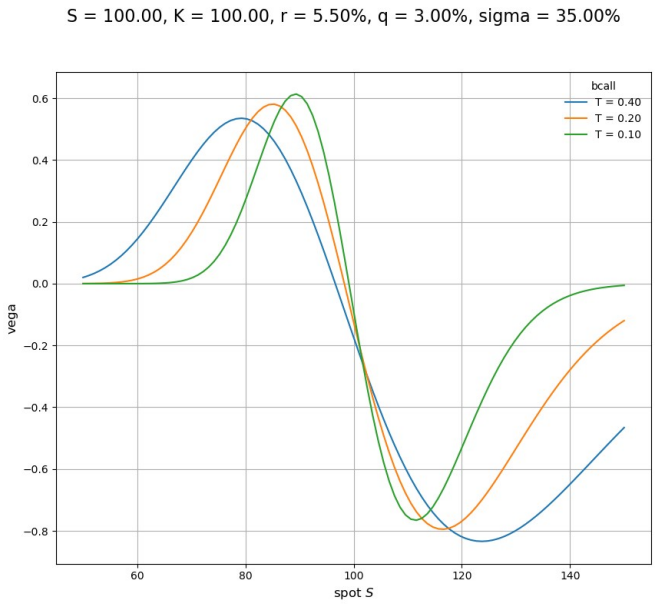
    → need to take into account volatility surface

- Large gamma, vega variation near barrier

  → (approximate) static hedging is preferred!







Barrier options:

- Standard single barrier options:

  Call options:

  Assume (t,T) is the life of the option, B is the barrier,

  - Knock-out (KO) call:
    - Up and out (UO) call ($B > K$):

      $C_{UO}(S_T,T,B) = \max(S_T - K, 0) I_{(0,B,t,T)}$, "reverse"

    - Down and out (DO) call:

      $C_{DO}(S_T,T) = \max(S_T - K, 0) I_{(B,\infty,t,T)}$, "regular"

  - Knock-in (KI) call:
    - Up and in (UI) call:

      $C_{UI}(S_T,T,B) = \max(S_T - K, 0)\left(1 - I_{(0,B,t,T)}\right)$, "regular"

    - Down and in (DI) call:

      $C_{DI}(S_T,T,B) = \max(S_T - K, 0)\left(1 - I_{(B,\infty,t,T)}\right)$, "reverse"

Closed form solutions may be obtained under constant parameters

- Parity:

  $$C_{KO}(S_t, t, B) + C_{KI}(S_t, t, B) = C(S_t, t)$$

- Barrier monitoring: e.g., daily close
- Rebate in KO options

Put options:

Similarly constructed ….

- Double barrier options:

  (UO, DO), (UO, DI), (UI, DO), (UI, DI)

- Type of barriers:
  - Continuous vs discrete monitored barriers
  - One-touch barriers
  - Parisian barriers
  - Asian (fixed or moving period of average) barriers
  - "m out of n days" barriers (common in convertible markets)
  - Out-side barriers (common in FX markets)
- Usage:
  - Cheaper alternatives of vanilla options
  - Yield generators:

    Reverse convertible notes (RCN): high yield bond + short DI put
- Risk:
  - "regular" vs "reverse"
  - Vega:
    - KO: less vol sensitivity than the corresponding vanilla
    - KI: more vol sensitivity than the corresponding vanilla
    - Effect of volatility skew: strike and barrier

- Effect of volatility term structure: barrier monitoring during the life of the option

→ need for taking into account the volatility surface

- Rapid changing greeks near the barrier

→ "barrier bending" :

Dealers may quote barrier options with a shifted barrier. The size of the shift depends on the competitiveness of the market, time to maturity etc

- (approximate) static hedging

Exercise:

Use the MC techniques developed in the previous lecture (Module 9) price a KO barrier option under flat volatility surface.

Asian options:

- Payoff:

  For $0 = t_0 < t_1 < t_2 < ... < t_n = T$ (averaging may take place over a sub-interval of the life of the option)

  - Average price call:

$$payoff = \max\left(\frac{1}{n}\sum_{i=1}^{n} S_{t_i} - K, 0\right)$$

  - Average price put:

$$payoff = \max\left(K - \frac{1}{n}\sum_{i=1}^{n} S_{t_i}, 0\right)$$

  - Average strike call:

$$payoff = \max\left(S_T - \frac{1}{n}\sum_{i=1}^{n} S_{t_i}, 0\right)$$

  - Average strike put:

$$payoff = \max\left(\frac{1}{n}\sum_{i=1}^{n} S_{t_i} - S_T, 0\right)$$

- Geometric average: $\left(\prod_{i=1}^{n} S_{t_i}\right)^{\frac{1}{n}}$

  e.g., geometric average price call = $\max\left(\left(\prod_{i=1}^{n} S_{t_i}\right)^{\frac{1}{n}} - K, 0\right)$, etc

  Closed form solution possible (under constant parameters)

  → Bound for arithmetic averaged options: $\left(\prod_{i=1}^{n} S_{t_i}\right)^{\frac{1}{n}} \leq \frac{1}{n}\sum_{i=1}^{n} S_{t_i}$

  → Serves as a control variate in Monte Carlo simulation

- Usage:
  - Popular in FX and commodity markets

    Effective hedging instruments for corporations with streams of liabilities
  - "accelerated share repurchase" (ASR)

Exercise:

Use the MC techniques developed in the previous lecture (Module 9) price an Asian option under flat volatility surface.

Lookback options:

- Payoff:

  Let $S^M = \max(S_{t'} : t \leq t' \leq T)$, $S^m = \min(S_{t'} : t \leq t' \leq T)$

  - Floating strike lookback call:

    $Payoff = \max(S_T - S^m, 0) = S_T - S^m$
  - Floating strike lookback put:

    $Payoff = \max(S^M - S_T, 0) = S^M - S_T$
  - Fixed strike lookback call:

    $Payoff = \max(S^M - K, 0)$
  - Fixed strike lookback put:

$$Payoff = \max(K - S^m, 0)$$

Observation frequency/period for determining S^M and S^m may vary.

→ Can be expensive

- Usage:

  Useful hedging instruments in FX and interest rate markets

Dispersion options:

So far in the course we have mostly focused on derivatives with single asset as underlying. In the case a stock index as the underlying the index (which is a basket of stocks) is treated as single asset paying continuous dividend yield. "Dispersion" explores differences in the asset performance.

Given a basket of assets $S^1, S^2, ..., S^n$

- Worst of/best of call/put:
  - Worst of call:

$$payoff = \max\left(\min(\frac{S_T^1}{S_0^1}, \frac{S_T^2}{S_0^2}, ..., \frac{S_T^n}{S_0^n}) - K, 0 \right)$$

  - Worst of put:

$$payoff = \max\left(K - \min(\frac{S_T^1}{S_0^1}, \frac{S_T^2}{S_0^2}, ..., \frac{S_T^n}{S_0^n}), 0 \right)$$

  - Best of call:

$$payoff = \max\left(\max(\frac{S_T^1}{S_0^1}, \frac{S_T^2}{S_0^2}, ..., \frac{S_T^n}{S_0^n}) - K, 0 \right)$$

  - Best of put:

$$payoff = \max\left(K - \max(\frac{S_T^1}{S_0^1}, \frac{S_T^2}{S_0^2}, ..., \frac{S_T^n}{S_0^n}), 0 \right)$$

- Rainbow options:

$$payoff = \max\left(\sum_{j=1}^{m} w_j \left\{ R_T^i \right\}_j, 0 \right) \ , \quad R_T^i = \frac{S_T^i}{S_0^i} - 1 \ , \quad \sum_{j=1}^{m} w_j = 1$$

$\left\{R_T^i\right\}_j$: j-th best return in $\left\{R_T^i\right\}$ (j = 1 is the best, j = 2 is 2nd best, etc)

- Spread options:

  $payoff = \max\left(R_T^2 - R_T^1 - K, 0\right)$

  K = 0: also called outperformance option or exchange option (have closed form solutions under constant parameters)

Volatility, correlation and volatility skew are important for pricing/risk managing those products.

Other Structured Products

Autocallables:

Structured notes. Here "callable" means a call option the note holders sold to the issuer to allow the issuer to redeem the structure if conditions meet. Such structures provide enhanced (but conditional) yield than conventional bonds.

- Payoffs:

  Let $0 = t_0 < t_1 < t_2 < ... < t_n = T$, $K_1 < K_2$, N = notional, m = coupon rate

  $coupon(t_i) = mNI_{K_1}\left(\frac{S_{t_i}}{S_0}\right)$, if not previously redeemed

  $redemption(t_i) = N$, $\frac{S_{t_i}}{S_0} > K_2$

  K_1 : coupon trigger, K_2 : redemption (call) trigger

  Variations:

  - Autocallable participating notes: potential high participation (if not called early) of upside while protecting principal
  - Autocallable without principal protection: e.g., embedding a short DI put to enhance yield
  - ….

Cliquet structures (also called ratchet options):

- Building block – forward start options:

  Forward start call:

  $$payoff = \max\left(\frac{S_{t_2}}{S_{t_1}} - K, 0\right), \quad t_0 < t_1 < t_2$$

  t_0 : valuation date; t_1 : strike fix date; t_2 : option maturity date

  → need be priced with forward implied volatility → exposure to forward skew

- A typical cliquet payoff:

  For $0 = t_0 < t_1 < t_2 < ... < t_n = T$, notional = 1,

  $$payoff = \sum_{i=1}^{n} \max\left(\min\left(\frac{S_{t_i}}{S_{t_{i-1}}} - 1, cap\right), floor\right)$$

  Valuation: need to model the dynamics of implied volatility surface properly

- Numerous variations in payoffs:
  - Variable local cap and floors, global cap and floors
  - Reverse cliquets:

    $$payoff = \max\left(M - \sum_{i=1}^{n} \max\left(1 - \frac{S_{t_i}}{S_{t_{i-1}}}, 0\right), 0\right)$$

  - Napoleons:

    $$payoff = \max\left(M + \min\left(\frac{S_{t_i}}{S_{t_{i-1}}} - 1 \middle| i = 1, ..., n\right), 0\right)$$

  - Digital cliquets
  - Cliquets with lock-in feature (accumulators)
  - Multi-assets
  - …..